

Intent Routing -- eBPF Kernel Programming Practical Guide

All steps completed on Ubuntu 24.04 with Minikube + Cilium environment

1. Environment Setup (Ubuntu Machine)

```
sudo apt update
sudo apt install -y \
  clang llvm libelf-dev libbpf-dev \
  gcc make iproute2 iputils-ping \
  linux-headers-$(uname -r)
```

2. Project Structure

```
cd ~
mkdir -p intent-routing/{src,policies,scripts,test}
cd intent-routing
touch README.md
```

3. Intent Policy File

```
{
  "intent": "low-latency",
  "threshold_ms": 100,
  "action": "reroute"
}
```

4. Simulate Intent Decision Logic

Python script to simulate decisions:

```
import json
import random
from pathlib import Path

policy_path = Path(__file__).parent.parent / "policies" / "latency-
intent.json"
with open(policy_path) as f:
    intent = json.load(f)

latency = random.randint(50, 200)
```

```
threshold = intent["threshold_ms"]

if latency > threshold:
    print(f"Latency {latency}ms > {threshold}ms → REROUTE triggered")
else:
    print(f"Latency {latency}ms OK → no action")
```

5. Generate Kernel BTF Header

```
cd src
sudo bpftool btf dump file /sys/kernel/btf/vmlinux format c > vmlinux.h
```

6. eBPF Kernel Program

```
#include "vmlinux.h"

typedef unsigned char    __u8;
typedef unsigned short   __u16;
typedef unsigned int     __u32;
typedef unsigned long long __u64;
typedef int              __s32;
typedef long long        __s64;

typedef __u16 __be16;
typedef __u32 __be32;
typedef __u32 __wsum;

#include <bpf/bpf_helpers.h>
#include <bpf/bpf_tracing.h>

char LICENSE[] SEC("license") = "GPL";

SEC("kprobe/tcp_v4_connect")
int BPF_KPROBE(bpf_prog)
{
    bpf_printk("Intercepted TCP connection\n");
    return 0;
}
```

7. Compile the eBPF Program

```
cd ~/intent-routing
clang -O2 -g -target bpf -D__TARGET_ARCH_x86 \
    -I./src \
    -c src/intent_router.c \
    -o intent_router.o
```

8. Install Working bpftool Binary

```
cd ~
curl -LO
https://github.com/libbpf/bpftool/releases/download/v7.5.0/bpftool-v7.5.0-
x86_64
chmod +x bpftool-v7.5.0-x86_64
sudo mv bpftool-v7.5.0-x86_64 /usr/local/bin/bpftool
hash -r
```

9. Load Program into Kernel

```
sudo rm -f /sys/fs/bpf/intent_router
sudo bpftool prog load intent_router.o /sys/fs/bpf/intent_router type
kprobe
```

10. Attach Program to Hook

```
sudo bpftool prog attach pinned /sys/fs/bpf/intent_router \
kprobe tcp_v4_connect
```

11. View Kernel Output

Terminal 1:

```
sudo cat /sys/kernel/debug/tracing/trace_pipe
```

Terminal 2:

```
curl https://example.com
curl https://google.com
```

Output (expected):

```
Intercepted TCP connection
Intercepted TCP connection
```